Center for Digital Transformation
TUM School of Management
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Exercise Sheet 1 – Economics of Digital Goods and Markets

Part 1: Enhancing understanding of concepts through application and repetition:

Question 1 – Digital vs. physical goods

Digital goods often require substantial upfront investment but can be reproduced and distributed at very low marginal cost. Physical goods, in contrast, typically incur additional costs for each unit produced. Which statement best describes the economic implications of these cost structures?

(a) Digital goods have no fixed costs but high marginal costs, whereas physical goods have constant total costs regardless of output.
(b) Digital goods have lower total costs than physical goods at every output level because their marginal costs are negligible.
(c) Digital goods have constant average costs as output increases, whereas physical goods always experience declining average costs.
(d) Digital goods can have high fixed costs but very low marginal costs, allowing average costs to decline as output increases.

Question 2 – Marginal vs. average cost

A digital textbook platform has the total cost function $TC(q) = 200{,}000 + 0.10 \cdot q$ (in €), where q is the number of downloads. At q = 500,000, what are the marginal cost (MC) and the average total cost (ATC) per download?

(a) MC = €0.10; ATC = €0.40
(b) MC = €0.50; ATC = €0.10
(c) MC = €0.10; ATC = €0.50
(d) MC = €0.40; ATC = €0.50

Question 3 – Pricing at marginal cost

Serving 20,000 users of a cloud-based tool costs €130,000 in total; serving 25,000 users costs €135,000 in total. What is the marginal cost per additional user, and what is the result at 25,000 users if the firm charges every user a price equal to that marginal cost?

(a) MC = €1.00; loss of €110,000
(b) MC = €1.00; break-even
(c) MC = €5.40; break-even
(d) MC = €0.20; loss of €110,000

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Question 4 – Platform launch challenges

Which pair best describes the two challenges during platform launch: chicken-and-egg problem and the penguin problem?

(a) Chicken-and-egg: users wait until others on their own side have adopted; penguin: sellers cannot join before consumers because of cross-side effects.

(b) Chicken-and-egg: each side of a two-sided platform joins only if the other side is already present; penguin: individual users wait for others to adopt first so they are not stranded on an empty platform.

(c) Both describe the high development cost the platform owner must bear before launch.

(d) Chicken-and-egg: the platform's prices are set too high for consumers; penguin: sellers use several competing platforms at once.

Question 5 – Launch tactics

A new online marketplace gives its first 500 sellers a €50 sign-up credit and runs onboarding workshops for them. Which adoption stimulus is this, and what is it meant to achieve?

(a) Seeding, Marquees: it permanently raises the platform's stand-alone value.

(b) Policy distinction: it raises the value of the platform for buyers before launch.

(c) Subsidizing, onboarding: it lowers the cost of joining until critical mass is reached.

(d) Subsidizing, onboarding: it increases economies of scale among sellers.

Question 6 – Launch tactics mechanisms

Digital platforms employ different launch tactics to overcome initial adoption barriers and reach critical mass. Some tactics increase the value of joining the platform, while others reduce the costs or effort associated with participation. Which of the following correctly classifies platform launch tactics as value-increasing or cost-decreasing?

(a) Value-increasing: Seeding, Marquees, Fringe benefits; Cost-decreasing: Onboarding, Knowledge transfer.

(b) Value-increasing: Onboarding, Knowledge transfer, Fringe benefits; Cost-decreasing: Seeding, Marquees.

(c) Value-increasing: Seeding, Knowledge transfer, Marquees; Cost-decreasing: Onboarding, Fringe benefits.

(d) Value-increasing: Marquees, Fringe benefits, Onboarding; Cost-decreasing: Seeding, Knowledge transfer.

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Part 2: Training ability to evaluate strategies and to create digital business model innovations:

Question 7 – Platform launch strategy: Start Small

A startup develops a new digital platform that connects freelance personal trainers with customers looking for individual fitness sessions. The platform aims to benefit from indirect network effects between trainers and customers but initially has limited financial resources and few users. Which of the following launch strategies would be most effective in overcoming the chicken-and-egg problem by applying the "Start Small" strategy?

(a) Launching the platform nationwide while investing heavily in advertising to attract as many trainers and customers as possible across different regions.

(b) Initially focusing on recruiting a large number of personal trainers across several major cities before actively marketing the platform to potential customers.

(c) Initially launching in a single city district, recruiting local personal trainers and attracting nearby customers to establish sufficient supply and demand before expanding geographically.

(d) Initially offering the platform exclusively to a small group of selected personal trainers, providing them with free tools before opening the platform to customers nationwide.

Question 8 – Platform launch strategy: Hell-bent

A startup develops a new digital marketplace connecting homeowners with local cleaning service providers. To overcome the chicken-and-egg problem, the startup raises substantial venture capital and decides to pursue a “Hell-bent” launch strategy, aiming to rapidly attract a critical mass of users on both sides of the platform. Which of the following approaches would be most effective in accelerating platform growth and establishing network effects?

(a) Investing heavily in nationwide advertising campaigns to build brand awareness among homeowners, while offering cleaning service providers free registration and reduced transaction fees to encourage participation.

(b) Offering substantial discounts on cleaning services to attract homeowners, while relying on the resulting increase in customer demand to organically attract cleaning service providers.

(c) Investing heavily in recruiting professional cleaning service providers through financial incentives, while using targeted digital advertising to gradually attract homeowners.

(d) Investing heavily in coordinated marketing campaigns targeting both homeowners and cleaning service providers, combining customer discounts, provider sign-up bonuses, and referral incentives to rapidly build a balanced user base on both sides of the marketplace.

Question 9 – Verification costs and margins

Hotel chains and licensed taxi operators have seen margins fall as platforms such as Airbnb and Uber spread. Which mechanism from the reduction in verification costs explains this best?

(a) Incumbents automate quality checks, cut their marginal cost and keep the saving as extra profit.

(b) Reputation and rating systems let unknown entrants credibly signal quality, which erodes the premium incumbents earned as trusted brands or licensed providers, so entrants compete on price.

(c) Buyers can verify quality cheaply, so they pay more for established brands as a quality signal.

(d) Firms track customers cheaply, which allows finer price discrimination and raises margins.

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Question 10 – Reputation system

Digital marketplace platforms such as eBay and Amazon reduce search costs, connecting buyers and sellers who often do not know each other. To facilitate transactions, platform owners need to reduce verification costs and establish trust between market participants. Which of the following measures is most effective in reducing verification costs for buyers?

(a) Allowing sellers to remove negative reviews from their profiles to maintain a positive reputation and encourage transactions.

(b) Implementing verified-purchase reviews, identity verification, and mechanisms to detect fraudulent ratings, thereby increasing the reliability of information about sellers and products.

(c) Allowing only sellers with established brands to join the platform, eliminating the need for customer reviews and other verification mechanisms.

(d) Increasing transaction fees for all sellers to signal higher quality, assuming that more expensive sellers are inherently more trustworthy.

Question 11 – Reputation system strategies

On digital marketplaces such as Amazon and eBay sellers rely on reputation systems to establish trust with potential buyers. Ratings and customer reviews influence purchasing decisions and can affect a seller's competitiveness. Which of the following strategies is most effective for a seller to build and maintain a strong reputation on a digital marketplace?

(a) Prioritizing the collection of a large number of positive customer reviews through targeted review requests, while focusing less on addressing individual complaints, since buyers primarily rely on average ratings.

(b) Focusing primarily on offering the lowest prices, since competitive pricing eliminates the importance of ratings and customer reviews.

(c) Consistently delivering high-quality products or services, providing accurate descriptions, and responding professionally to negative feedback.

(d) Focusing on maintaining high average ratings by concentrating on short-term customer satisfaction, rather than investing in long-term service reliability and consistent product quality.

Question 12 – From economics to subscription access business model

A music streaming service charges a flat €10 per month for unlimited access to its catalogue. Assume content licensing is a fixed annual fee. Which statement best explains why this subscription model is economically viable?

(a) Per-play pricing would be optimal because each play is costly to serve, so the flat fee leads to structural losses.

(b) The flat fee works because plays are rival: each play uses up the song for other listeners, which creates scarcity.

(c) The flat fee works because fixed costs are zero, so every euro of subscription revenue is profit.

(d) Near-zero marginal cost per additional stream makes usage-based pricing unnecessary; the flat fee recovers the high fixed content costs from many subscribers, but the model depends on reaching sufficient scale.

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Question 13 –Bundling and subscription pricing

Digital content platforms such as Spotify offer consumers access to millions of songs for a single monthly subscription price. Because consumers have very different preferences and willingness to pay for individual songs, digital content platforms bundle the content together into one product and charge a single subscription price. Which of the following best explains why bundling facilitates charging a single subscription price?

(a) Bundling reduces differences in consumers' preferences for individual songs, making their willingness to pay for each song more similar and allowing the platform to charge a uniform subscription price.

(b) Bundling allows platforms to offer millions of songs at a single price because the marginal costs of digital goods are low, making consumers' willingness to pay largely irrelevant for subscription pricing.

(c) Bundling combines many digital goods whose individual valuations differ across consumers. As these differences tend to average out across a large bundle, consumers' total willingness to pay becomes more predictable, making a single subscription price more effective.

(d) Bundling increases consumers' willingness to pay by ensuring that every subscriber values all songs in the library equally, allowing the platform to charge a subscription price that captures the full value of its content.

Question 14 – Subscription pricing

Digital content platforms such as Netflix and Spotify provide access to extensive libraries of movies, series, or music for a single monthly subscription price. This pricing model is particularly attractive because digital goods have low marginal costs and consumers' heterogeneous preferences tend to average out across large content bundles. Why does offering a large variety of digital content at a single subscription price create value for digital platforms?

(a) It allows platforms to capture a larger share of consumers' willingness to pay because individual differences in content valuations tend to average out across large bundles.

(b) It enables platforms to offer flat-rate access to extensive content libraries, eliminating the need to charge consumers separately for each movie, series, or song.

(c) It allows platforms to generate recurring subscription revenues, making revenues more predictable and reducing their dependence on individual content transactions.

(d) All of the above.

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Question 15 – Subscription pricing and value extraction

A digital music platform offers two songs, Pop and Rock, to four potential customers. Each customer has a different willingness to pay (WTP) for the individual songs, as shown in the table below.

	Consumer A	Consumer B	Consumer C	Consumer D
Pop	€10	€2	€8	€4
Rock	€2	€10	€4	€8
Total WTP	€12	€12	€12	€12

The platform considers two pricing strategies:

- Individual pricing: Each song is sold separately at a uniform price for all consumers. The platform can set different prices for Pop and Rock.
- Pure bundling: Both songs are offered together at a single bundle price.

Assume that consumers purchase a song or bundle whenever their willingness to pay is at least as high as its price, and that marginal costs are zero. Which of the following statements correctly compares the maximum profit under individual pricing and pure bundling?

(a) Individual pricing generates €40, while bundling generates €48, because bundling eliminates differences in consumers' preferences for individual songs.

(b) Individual pricing generates €32, while bundling generates €48, because bundling combines heterogeneous individual valuations into identical total valuations, allowing the platform to capture more consumer value.

(c) Individual pricing generates €48, while bundling also generates €48, because the sum of consumers' willingness to pay is independent of the pricing strategy.

(d) Individual pricing generates €36, while bundling generates €40, because charging a single price requires the platform to offer a discount compared with the sum of individual valuations.

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